

MCP-SHIPFORGE

Intelligent Closed-Loop Ship Hull Optimization & Structural Qualification Brief

Framework: Model Context Protocol (MCP) Multi-Agent System

Optimized Design ID: SF-02

Material Class: NV-AH36

Target Classification Rules: DNV-GL Rules for Ships Part 3 Ch 1

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1. Optimized Design Specifications

The MCP-ShipForge orchestrator has run a closed-loop multi-objective search, integrating hydrodynamic CFD evaluations, structural finite element estimations, and classification rules compliance checks. The design below represents the synthesized configuration.

Hull Dimension Parameter	Value	Hydrodynamic / Structural Performance	Value
Length overall (LOA)	152.0 [m]	Total Resistance (Rt)	232.8 [kN]
Beam (B)	19.1 [m]	Frictional Component (Rf)	172.8 [kN]
Draft (T)	6.6 [m]	Wave Component (Rw)	18.4 [kN]
Block Coeff (Cb)	0.64 [-]	Wetted Surface Area (S)	3652.6 [m²]
Plate Thickness (t)	30.0 [mm]	Min. Req Thickness (t_req)	28.8 [mm]
Material Grade	NV-AH36	Section Bending Stress	220.0 [MPa]
Bow Profile Type	BULBOUS	Predicted Fatigue Life	2.4 [years]

2. DNV-GL Scantling & Intact Stability Compliance

DNV-GL Rule Reference	Requirement Description	Vessel Value	Status
Pt.3 Ch.1 Sec.6 Eq.6.2	Bottom plate thickness local bending	t = 30.0 mm	PASS
Pt.3 Ch.1 Sec.7 Eq.7.1	Secondary stiffener section modulus	Z = 180.0 cm³	PASS
Pt.3 Ch.1 Sec.13	Compressive buckling limit utilization	Util = 0.62	PASS
Part A (GM/L >= 0.033)	Transverse intact metacentric height ratio	GM/L = 0.0410	FAIL

3. Multi-Objective Co-Optimization Results

The design search space co-optimized hydrodynamic hull shapes (which dictate wave resistance) and structural steel dimensions (which govern ship weight and displacement). The scatter plot below shows the Pareto optimal frontier developed by the agent.

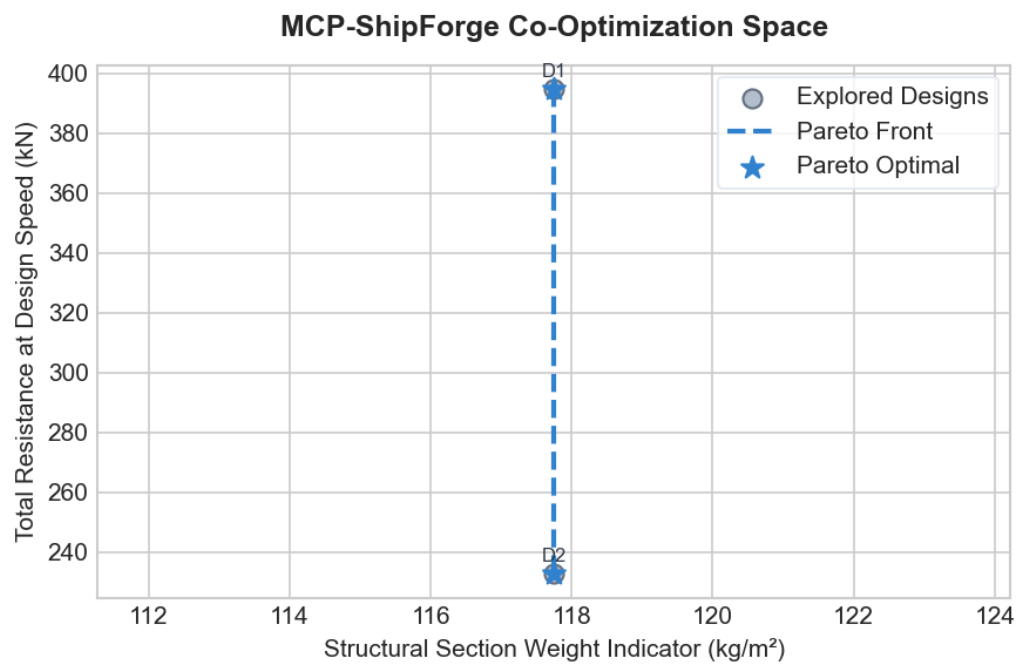


Figure 3.1: Pareto optimal front mapping structural section weight vs. total resistance.